

PET final-design selection study

Study deck (generated from committed results)

PET final-design study

PET is diagnostic method development. Nothing here is a publication adoption; no historical threshold, verdict or the predecessor campaign's disposition changes; simulation only.

What the evidence supports

question	what the evidence supports
Can improved PET recover meaningful physics changes? Is it more accurate than GBDT OmniFold?	Yes, in this simulation test. H2S1T24 $K=5$ passed every frozen point-estimator rule: energy tilt 0.895, proton topology 0.391, energy $\times q_3$ 0.446. On energy shape, yes; on generator changes, no. On the same final-bank events PET beats a tuned scalar GBDT OmniFold on the energy tilt (H2S1T24 +0.102, L128S1T24 +0.070 in R) but is worse on every generator-model reweighting (NuWro -0.247, GiBUU -0.206 for H2S1T24). Binned IBU (0.923 on the development tilt) matches PET on the 1D tilt but fails the proton case.
Can it give tighter, reliable confidence intervals?	Unestablished. Its six-member bootstrap interval failed calibration: too wide overall (68 % intervals cover 0.896), too narrow at low acceptance (0.257).
Will it distinguish generators more strongly?	Unestablished. That needs calibrated inference including detector and model uncertainties; no tested procedure provides it, and PET recovers generator-model changes less well than GBDT.

Terminal outcome by the frozen rules: **NO_ELIGIBLE_DESIGN**, no selected default. Signal-only simulation, conditional on the frozen event banks; PET is diagnostic.

The two finalists side by side

quantity (final bank, look 1)	H2S1T24 $K=5$	L128S1T24 $K=4$
Energy-tilt recovery (E0)	0.895	0.863
Proton-topology recovery (E4)	0.391	0.348
Joint energy-momentum-transfer recovery (E5)	0.446	0.450
Estimator-seed spread of energy recovery (N2)	0.037	0.043
Largest observed truth weight (N1)	90.2	33.4
Measured A100-hours per unfolding	1.77	1.52
Frozen point-estimator rules	all pass	B2 fails
Interval calibration (coverage)	fails (C1, C4)	not tested (ineligible on B2)

Recoveries: 1 = exact, 0 = no better than the prior. B2's failure is a point decision whose interval crosses the limit; the two designs are not shown to differ in robustness.

Design names

name part	meaning
CTL	historical production PET recipe: misses carried, 3 iterations, 8 epochs, forced low rate after iteration 0
C	CTL with the predecessor's efficiency-corrected truth step
B	predecessor's feature arm: reco energy summaries at step 1, one-hot particle type at step 2
H1	C + categorical truth particle type (13-class one-hot) in the truth step
H2	H1 + reco summaries at the detector step (reco E_{avail} , reco q_3 , cluster ΣE and count)
S1	constant learning rate at every iteration
E16 / T24	16 epochs per fit at both steps / 24 epochs per truth-step fit
L64 / L128	wider detector-step PET (projection 64 / 128, 4 layers)
P2pre / P2scr	Gregor's PET2-small backbone at the detector step on its 33-token typed inputs, pretrained / from scratch
A1	annealed detector-step learning rate after iteration 1
X4	4 independently seeded truth-step fits averaged per iteration
$K=5$, $K=4$	the frozen number of OmniFold iterations

Names compose: $H2S1T24 = H2 + S1 + T24$. Step 1 reweights simulation to match (pseudo)data at detector level; step 2 turns those weights into truth-level weights; one pass is one iteration.

How the numbers are measured

- ▶ **Simulation only.** Pseudodata are simulated events with a known, injected change to the truth (e.g. an E_{avail} tilt, more protons); the unfolding starts from the unchanged simulation (the prior).
- ▶ **Recovery** $R = 1 - L_1(\text{unfolded} - \text{target}) / L_1(\text{prior} - \text{target})$ on normalized histograms: 1 = the injected change fully recovered, 0 = no better than the prior, < 0 = moved away.
- ▶ **Tests:** energy tilt and its opposite (E0, E3), proton topology (E4, $E_{\text{avail}} \times$ proton class), energy $\times$ momentum transfer (E5), a 21-case robustness library, weight tails and seed spread.
- ▶ **Fresh events:** designs were developed on a DEV bank; the final test used a bank no earlier work had scored, blinded until every final run was complete. A third bank stays sealed.
- ▶ **Rules frozen first:** every threshold, finalist and sample size was committed before final results were seen; the outcome is whatever those rules give.
- ▶ **Coverage:** does the six-member bootstrap interval contain the true value at its nominal rate across 120 independent replicates?

Question, preference and decision rule

- ▶ Select one complete PET design (detector/truth representations, model and initialization, recipe, miss rule, iteration count, uncertainty procedure) among a declared, tested family — or conclude “no eligible design” / “unresolved” with the discriminating evidence.
- ▶ The smaller network is preferred *only* if non-inferior (simultaneous one-sided 95 %, margin -0.02 on the primary recovery, plus frozen regional/topology margins) *and* at least $2\times$ cheaper in measured total cost; otherwise the stronger larger/pretrained design.
- ▶ Numeric decision table frozen before any successor run (PROTOCOL-20260925.md); final inference conditional on frozen event banks; final-bank runs blinded until an explicit UNBLIND amendment.

Fresh-event capacity and independence

bank	rows	use
DEV (exposed)	45089191	development; every prior
FB (never drawn)	2646891	final pseudodata; sealed except listed runs
RB (pool R, never opened)	1414846	reserve; sealed

Only ≈ 3.4 historical-size replicates of never-drawn events exist and no further same-model simulation: final replicates are independent random subsets *given the banks*; the bank's own offset is reported beside every contrast.

Where the multiplicity failure sits: the detector step

Injected E_{avail} L1: tilt 0.273, protons $\times$ 1.3 0.026, neutrons $\times$ 1.3 0.011 — the multiplicity cases barely move the scored marginal.

	det. ΣE	$k=1$	$k=3$	$k=10$	pull p -class	$k=1$	$k=3$	$k=10$
CTL		-0.16	-0.37	-0.39		+0.01	+0.02	+0.05
C		-0.16	-0.45	-0.44		+0.01	+0.03	+0.08
B		+0.81	+0.82	+0.83		+0.11	+0.22	+0.49

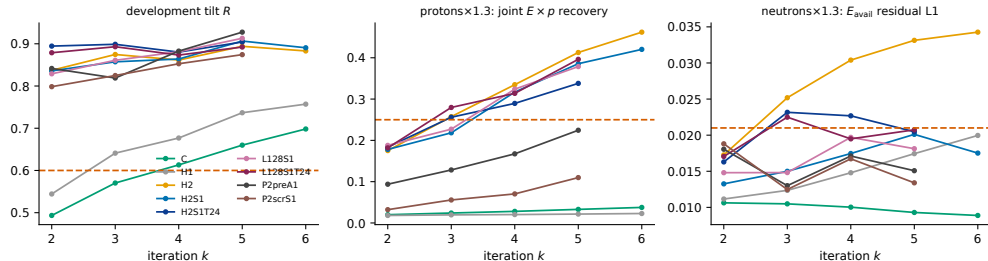
Protons $\times$ 1.3 (2 replicates): the baseline detector step (CTL, C) mis-closes the cluster-energy distribution and transmits almost no proton-content change to the pull; with reco summaries (B) both are transmitted. Development evidence.

Truth-step learnability (bounded interventions)

truth arm (step 2 on the exact weights)	p-class sel.	p-class missed	n-class sel.
raw PDG	0.83	0.83	0.59
one-hot PDG	0.87	0.88	0.88
one-hot + counts	0.89	0.92	0.88
one-hot, 24 epochs	0.97	0.96	0.98

Given C's real iteration-1 pull, the one-hot truth step recovers 0.012 of the joint $E_{\text{avail}} \times \text{proton change}$; given B's, 0.114. The truth step learns and extrapolates what it receives; the loss is upstream.

Development screen (DEV bank)



Reco summaries lift the tilt (H2S1 $k=5$: 0.906; C $k=3$: 0.570); the forced-1e-5 schedule fails the neutron screen (H2 $k=3$: 0.025) while the constant rate passes (H2S1 $k=5$: 0.020, topology 0.386). Development evidence, 2 draws per case.

Reproducibility repair (single-unfolding seed spread)

design	truth step	k	seed sd of R_{E0}	N2 (≤ 0.05)
H2S1	8 epochs	5	0.095	fail
L128S1	8 epochs	5	0.091	fail
L64S1	8 epochs	4	0.057	fail
CS1	8 epochs	6	0.054	fail
H2S1E16	16 epochs (both steps)	4	0.074	fail
L128S1E16	16 epochs (both steps)	4	0.071	fail
H2S1T24	24 epochs	5	0.037	pass
L128S1T24	24 epochs	4	0.043	pass

DEV draws $0-1 \times 4$ estimator seeds at fixed events. The spread sits in the truth step (the detector step varies little with the seed); a 24-epoch truth step repairs it (Amendment 3b).

Finalist rule applied mechanically

design	K^*	dev. tilt at K^*	S-N1	seed sd ($S-N2 \leq 0.05$)
H2S1T24	5	0.904	PASS	0.037
H2S1	5	0.906	PASS	0.095
H2S1E16	4	0.881	PASS	0.074
CS1	6	0.680	PASS	0.054
L128S1T24	4	0.873	PASS	0.043
L128S1	5	0.913	PASS	0.091
L128S1E16	4	0.875	PASS	0.071
L64S1	4	0.882	PASS	0.057
P2preS1	4	0.905	fail	—
P2preA1	—	—	—	—
P2scrS1	—	—	—	—

Rule committed before the recoveries it ranks were inspected, with addenda (completeness, S-N1, S-N2); finalists: compact H2S1T24 $K=5$, large L128S1T24 $K=4$ (Amendments 3c, 3d).

Algorithmic alternative: AUSSIE (scalar, matched)

AUSSIE setting vs OmniFold $k=3$	moves-away: AUSSIE	OmniFold	worst R : AUSSIE	OmniFold
$\lambda=1000$, misses carried, kinematics	11	6	-1.077	-0.217
$\lambda=1000$, misses carried, kinematics + species	8	6	-0.985	-0.230
$\lambda=0$, efficiency-corrected, kinematics	14	6	-3.876	-0.392
$\lambda=0$, efficiency-corrected, kinematics + species	7	6	-0.735	-0.352

Bounded matched stress test (same scalar inputs, miss handling, data and tuning opportunity); truth kinematics = $(E_{\text{avail}}, p_T, p_{\parallel}, q_3)$. 12 comparisons: 2 miss rules $\times$ 2 truth-input sets $\times$ 3 OmniFold operating points (shown: $k=3$). AUSSIE wins none of the 12 on robustness or stability, so it does not advance to a PET backbone.

Frozen finalists and final-stage design

- ▶ Compact **H2S1T24** ($K=5$) and large **L128S1T24** ($K=4$): detector reco summaries, categorical truth PDG, constant rate, 24-epoch truth step; step-1 PET 47 k vs 0.97 M parameters. Anchors CTL and C at $k=3$.
- ▶ Cost per unfolding (charged A100-h, declared packing): 1.77 (compact) vs 1.49 (large): the smaller network is not cheaper, so the cost-based preference cannot apply.
- ▶ Independent sizing: $n_F = 60$ (capped; E_0 non-inferiority between the finalists would need 250 draws — a quantified limit); D4c/D3 draws 40.
- ▶ Final bank: FINAL, 21-case library and coverage run blinded until the UNBLIND amendment.

Look 1: finalists on the final bank

look 1 (mean, simultaneous LB)	H2S1T24 $K=5$	L128S1T24 $K=4$
E_0 development tilt (U1)	0.895 (0.876)	0.863 (0.842)
moderate / good region (U2b)	0.874 / 0.917	0.846 / 0.910
E_3 opposite tilt (U3)	0.835 (0.815)	0.829 (0.809)
E_4 proton topology (U4)	0.391 (0.368)	0.348 (0.326)
$E_5 E_{\text{avail}} \times q_3$ (U5)	0.446 (0.429)	0.450 (0.435)
N2 seed sd (≤ 0.05)	0.037	0.043
gain over CTL in R_{E_0}	+0.589 (0.568)	+0.557 (0.535)
cost per unfolding (A100-h)	1.77	1.52
status (rules)	INCOMPLETE (C1-C5)	INELIGIBLE (B2)

Ranking: CONTINUE (candidates pending: H2S1T24K5). No look 2 (no sequential rule returned CONTINUE). Coverage of H2S1T24 runs blinded.

Look 1: what the verdict does and does not show

- ▶ **B2** (D4d n down, mean residual — injected E_{avail} L1, limit 0.010, a point rule): H2S1T24 0.0097 [0.0025, 0.0169] PASS; L128S1T24 0.0122 [0.0064, 0.0181] FAIL. **Point-decided, statistically unresolved**: both intervals cross the limit; the difference is not resolved by this comparison. Not evidence that the compact design is more robust, nor that the two are equivalent.
- ▶ **B1** PASSes under its frozen independence-based bound. Under within-draw dependence the evidence is insufficient to establish a per-unit failure probability ≤ 0.10 : on the common panel (FB0–7) H2S1T24 has 2/8 draws with a failure (CP upper 0.651), L128S1T24 1/8; with none the bound would be 0.369. This does not show the probability exceeds 0.10.
- ▶ Like-for-like and FB-population recoveries agree on the designated endpoints (report §6, report only).

Uncertainty calibration: coverage of H2S1T24

H2S1T24 $K=5$, $B = 6$, 120 replicates	measured	verdict
C1 pooled 95 % (LB ≥ 0.90 , point ≤ 0.99)	0.990 (LB 0.980)	unresolved
C1 pooled 68 % (LB ≥ 0.60 , point ≤ 0.80)	0.896 (LB 0.858)	decisive
C1 (both levels)		FAIL
C2 every bin 95 % ≥ 0.85	all bins pass	PASS
C3 moderate / good 95 % LB ≥ 0.85	0.949 / 0.993	PASS
C4 half-width / limit, bins 1–6 (≤ 1)	1.32–1.72	FAIL
C5 D4c up	skipped after decisive C1–C4 FAIL	—

In aggregate the six-member interval is **too wide**; the same against the FB population target. Not uniformly conservative: the low-acceptance region **under-covers** (68 %: 0.26, 95 %: 0.79), a region no rule gates. Mechanism not established.

PET vs GBDT OmniFold on the same events

paired difference in R , PET – GBDT	H2S1T24 $K=5$	L128S1T24 $K=4$
energy tilt E0 (60 draws)	+0.102 [0.084, 0.119]	+0.070 [0.051, 0.089]
opposite tilt E3 (60)	+0.076 [0.059, 0.093]	+0.070 [0.053, 0.088]
proton topology E4 (40)	+0.055 [0.035, 0.075]	+0.012 [−0.007, 0.031]
E4 against GBDT run to $k=10$	+0.009 [−0.012, 0.030]	−0.034 [−0.054, −0.015]
energy $\times q_3$ E5 (40)	+0.009 [−0.009, 0.026]	+0.013 [−0.005, 0.030]
generator reweighting: NuWro (8)	−0.247 [−0.345, −0.148]	−0.208 [−0.324, −0.092]
NuWro' (8)	−0.216 [−0.303, −0.130]	−0.180 [−0.241, −0.118]
GiBUU (8)	−0.206 [−0.305, −0.107]	−0.121 [−0.177, −0.065]

same draws	H2S1T24	L128S1T24	GBDT
B2 residual, D4d n down (PET rule ≤ 0.010)	0.0097	0.0122	0.0033
largest truth weight	90.2	33.4	5.6
GBDT's E0 summed per-bin MSE relative to PET	$2.9\times$	$1.9\times$	1

GBDT = the matched study's tuned scalar OmniFold (HistGradientBoosting, efficiency-corrected, reco summaries + muon kinematics, truth with species counts, $k=7$ chosen on DEV), run on the same look-1 final-bank units with the same events, targets and scorer. Unadjusted 95% intervals; exploratory. GBDT error is mostly systematic under-correction. No uncertainty comparison: this GBDT has no interval procedure here. On DEV, binned IBU and AUSSIE match PET on the 1D tilt and fail hadron or generator cases.

Terminal outcome (frozen rules)

NO_ELIGIBLE_DESIGN

finalist	status (rules)
H2S1T24K5	INELIGIBLE (C1, C4)
L128S1T24K4	INELIGIBLE (B2)

- ▶ No design selected and no default named (the default must itself be eligible).
- ▶ Supported: H2S1T24's point estimator passes every accuracy, robustness and stability rule; its interval procedure fails calibration (too wide in aggregate; under-covers at low acceptance). Not supported: a ranking of the finalists.
- ▶ Remaining route (owner decision): \$11 repair — interval calibrated on DEV, re-validated on the sealed RB bank.

Architectures tested, what came from Gregor's work, and why none passed

design	change	from Gregor's work	why it did not pass
H1	C + categorical truth PDG (13-way one-hot)	type encoding (cmp 2), padding kept distinct; truth step is ours	S-U4 fails at every k (best 0.023; need ≥ 0.25)
H2	H1 + reco summaries at step 1	whole-event energy sums + overflow (cmp 4, 6; not per-family)	no k passes all four screens
CS1	C + constant rate	—	S-N2 seed sd 0.054 > 0.05
H2S1	H2 + constant rate	as H2	S-N2 seed sd 0.095 > 0.05
H2S1E16	H2S1, 16 epochs	as H2	S-N2 seed sd 0.074 > 0.05
L64H2	H2, step-1 PET width 64	as H2 + one capacity step (cmp 7)	no k passes all four screens
L128H2	H2, step-1 PET width 128	as H2 + one capacity step (cmp 7)	no k passes all four screens
L64S1	H2S1, width 64	as L64H2	S-N2 seed sd 0.057 > 0.05
L128S1	H2S1, width 128	as L128H2	S-N2 seed sd 0.091 > 0.05
L128S1E16	L128S1, 16 epochs	as L128H2	S-N2 seed sd 0.071 > 0.05
P2preS1	PET2-small at step 1, pretrained	his backbone, checkpoint, 33-token typed inputs, recipe	S-N1 step-1 weight tail at $K^* = 4$
P2scrS1	PET2-small, random init	as P2preS1 without the checkpoint	S-U4 fails at every k (best 0.110; need ≥ 0.25)
P2preA1	P2preS1, annealed step-1 rate	as P2preS1	S-U4 fails at every k (best 0.225; need ≥ 0.25)
P2scrA1	P2scrS1, annealed step-1 rate	as P2scrS1	S-U4 fails at every k (best 0.060; need ≥ 0.25)
H2S1T24	H2S1, 24-epoch truth step (finalist)	as H2	final bank: coverage C1, C4 FAIL — INELIGIBLE (C1, C4)
L128S1T24	L128S1, 24-epoch truth step (finalist)	as L128H2	final bank: B2 FAIL (point-decided) — INELIGIBLE (B2)

Also closed: the step-2 ensemble fallback (H2S1X4, L128S1X4; entry condition never met) and AUSSIE (bounded matched scalar test). Anchors/controls (CTL, C, B) are not selectable. Development screens: S-U1 tilt, S-U3 opposite tilt, S-U4 proton topology, S-B2 D4d residual; then S-N1, S-N2. Gregor-derived parts follow the pinned comparison (cmp sections); the truth-step design is the study's own.

Reference methods and alternatives

method	what it is	from Gregor's work	result
CTL $K=3$	historical production PET recipe (anchor)	—	final bank energy tilt 0.306 (floor 0.556)
C $K=3$	efficiency-corrected truth step (anchor)	—	final bank energy tilt 0.560
B	predecessor's feature arm (control)	energy sums (cmp 4)	development tilt 0.474 at $k=3$
Gregor's arm	his PET2-small, his tokens, pretrained; historical comparison, $k=3$	his configuration (adapted)	0.417 vs ours 0.304; floor 0.556: NEITHER ELIGIBLE
GBDT OmniFold	scalar HistGradientBoosting on reco p_T , $p_{ }$, E_{avail} , q_3 , cluster sums	—	development tilt 0.783 (efficiency-corrected) / 0.418 (misses carried), $k=3$; proton topology 0.389 only with species-count truth inputs
binned IBU	response-matrix unfolding, efficiency-corrected	—	development tilt 0.923; moves away in the proton case (-0.381)
AUSSIE	scalar method of arXiv:2602.24282, matched to GBDT OmniFold	—	lost all 12 matched comparisons on robustness and stability

Anchors are not selectable; scalar references use the predecessor's DEV-bank selections, so they are context, not matched comparisons with the PET finalists.

What this study cannot authorize

- ▶ No publication adoption; no real-data unfolding; no $C_{\text{stat}}/C_{\text{ML}}$; no Gate-6 work; 0I-126 not reopened; the scalar-5D covariance unchanged; no note/primer/paper change.
- ▶ Historical thresholds and verdicts (`NEITHER_ELIGIBLE` / `NO_SELECTION`) and the predecessor's disposition are unchanged.
- ▶ Scope of any selection: signal-only simulation, the inventory's simulated detector response, the study's event scale, conditional on the frozen banks; generator dependence beyond histogram reweightings is unprobed.